



Evaluating Influenza Vaccine Effectiveness Against Mortality: A Causal Inference Approach

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Abstract:

Background: Seasonal influenza is a significant global health concern, contributing to considerable morbidity and mortality, especially among vulnerable populations such as the elderly and those with chronic conditions. While influenza vaccination is recognized as a crucial preventive strategy, its real-world effectiveness in reducing mortality, particularly among hospitalized patients, remains debated. This study examines the impact of influenza vaccination on survival outcomes among hospitalized patients in Australia between 2010 and 2017, focusing on differences across H1N1- and H3N2-dominated seasons.

Methods: A retrospective observational cohort of 1482 hospitalized patients with laboratory-confirmed influenza was analyzed. The primary outcome was survival during hospitalization. Vaccine effectiveness was assessed using a causal inference framework, including propensity score matching, optimal matching, and inverse probability of treatment weighting (IPTW) to adjust for confounders such as age, comorbidities, and functional status. Kaplan-Meier survival curves and Cox proportional hazards models were employed to estimate adjusted survival probabilities and hazard ratios. Subtype-specific analyses for H1N1 and H3N2 seasons were also conducted.

Results: Crude analyses revealed a paradoxical association, with vaccinated individuals exhibiting lower survival rates compared to non-vaccinated individuals ($p < 0.0001$). Adjusted analyses showed a neutral effect of vaccination, with no statistically significant difference in survival between groups after accounting for confounders. Subtype-specific analyses also revealed no significant survival benefit associated with vaccination during H1N1 or H3N2 seasons, with estimated average treatment effects near zero. Persistent imbalances in certain confounders, such as age and chronic respiratory disease, highlight the challenges of achieving complete adjustment.

Conclusions: This study underscores the critical role of confounding in interpreting the relationship between influenza vaccination and mortality. While vaccination appeared neutral in its impact after adjustments, the potential for unmeasured confounding and vaccine mismatch warrants cautious interpretation. Efforts to enhance vaccine effectiveness, particularly for high-risk populations, remain essential to reducing influenza-associated mortality. Future research should prioritize randomized trials or improved observational designs to strengthen causal inference.

1. Introduction

Seasonal influenza is a major global health concern, causing millions of cases of illness and hundreds of thousands of deaths each year. Vulnerable populations, including the elderly, pregnant individuals, and those with chronic conditions, are particularly affected, contributing significantly to the annual mortality and healthcare burden [1]. Vaccination is widely acknowledged as the most effective strategy for preventing influenza-related complications. However, the real-world effectiveness of influenza vaccines (VE), particularly in reducing mortality, remains a subject of intense investigation and debate [2].

This study examines the effectiveness of influenza vaccination in reducing influenza-related mortality among hospitalized patients in Australia from 2010 to 2017, while exploring whether vaccine impact varies across years dominated by H1N1 and H3N2 subtypes. Previous research has shown that the effectiveness of influenza vaccination can fluctuate significantly depending on the dominant circulating subtype and the degree of vaccine-virus antigenic match. Notably, H3N2-dominated seasons are often associated with higher mortality and morbidity than H1N1 seasons, due to greater genetic drift and reduced vaccine effectiveness in elderly populations [3,4].

Recent studies emphasize the importance of repeated annual vaccination to mitigate the effects of waning immunity and provide protection against emerging strains. However, evidence suggests that repeated vaccinations may also lead to inconsistent outcomes, particularly among older adults, where immunosenescence and antigenic interference may diminish vaccine-induced protection [4,5]. These findings highlight the need to better understand how influenza vaccines perform in diverse patient populations and across different epidemic seasons.

Despite the widespread implementation of vaccination programs, significant uncertainties remain regarding the magnitude of VE in reducing influenza-related mortality. For example, some studies report a strong protective effect of vaccination on reducing all-cause mortality, particularly among older adults with comorbidities, while others highlight methodological challenges such as selection bias and confounding by health status [6]. Furthermore, challenges in estimating VE are compounded by factors such as waning immunity, heterogeneity in host response, and differences in outcomes measured, including mortality, hospitalization, and severe illness [2,5].

To address these gaps, this study leverages a causal inference framework to evaluate the real-world impact of influenza vaccination on mortality, accounting for confounders and heterogeneity in patient characteristics. The findings aim to provide evidence for optimizing vaccination strategies, particularly in seasons with varying dominant subtypes and levels of vaccine-virus match. By focusing on mortality, the most critical influenza-related outcome, this research contributes to the ongoing efforts to enhance public health interventions and reduce the global burden of seasonal influenza.

2. Materials and Methods

2.1 Study Design and Population

This study utilized data from a cohort of 1482 patients hospitalized with laboratory-confirmed influenza across 17 Australian hospitals between 2010 and 2017. The retrospective, observational design allowed for the examination of vaccine effectiveness and influenza-related outcomes over a significant period, covering variations in influenza subtypes. Patients were categorized based on their vaccination status, with individuals considered vaccinated if they had received the seasonal influenza vaccine at least 14 days prior to hospital admission. This threshold was chosen to ensure sufficient time for the vaccine to elicit an immune response.

The primary outcomes of interest were influenza-related survival during hospitalization and the differential impact of influenza vaccination on mortality across seasons dominated by different influenza A subtypes, H1N1 and H3N2. These subtypes were identified as dominant based on surveillance data for the corresponding years. This dual focus enabled the assessment of vaccine performance not only on overall survival but also on its variability linked to the genetic and epidemiological characteristics of circulating strains.

2.2 Data Collected

Comprehensive data were collected to ensure a detailed characterization of the patient cohort and to capture the broad range of factors potentially influencing mortality. Demographic data included patient age, sex, and Indigenous Australian status. Clinical data encompassed smoking status and obesity, both of which are well-established risk factors for respiratory infections and complications.

In addition to these core variables, the dataset included information on pregnancy, homelessness, and nursing home residency. These categories were considered because of their relevance in understanding unique vulnerabilities to influenza and associated mortality. However, due to the low frequency of observations in these groups, these variables were excluded from the final analysis as they lacked sufficient statistical power to draw meaningful conclusions.

Detailed records of comorbidities were maintained, capturing conditions such as diabetes; chronic cardiac, respiratory, neurological, renal, and hepatic diseases; malignancies; and immunosuppressive conditions. These factors were considered critical confounders due to their established associations with higher risks of influenza complications.

Functional status was systematically assessed and categorized into severity levels—ranging from "Not Restricted" to "Bed-Bound"—to reflect the patients' overall health and independence at the time of admission.

2.3 Descriptive Statistics

The statistical analysis began with descriptive summaries to characterize the cohort. Categorical variables were analyzed in terms of absolute and relative frequencies, while continuous variables were summarized using means and standard deviations. The grouping into vaccinated and unvaccinated cohorts and further stratification by influenza subtype dominance (H1N1 vs. H3N2) allowed for detailed subgroup comparisons.

A multivariate analysis was performed to explore the relationships between key variables and survival outcomes, as well as to identify differences between vaccinated and unvaccinated groups. Continuous variables, such as age, were analyzed using t-tests to compare mean differences across groups, providing insight into whether age significantly varied between survivors and non-survivors or between vaccinated and unvaccinated individuals. Categorical variables were examined using chi-square tests to determine their association with survival and vaccination status. This

analysis aimed to identify which variables were significantly related to survival or vaccination, helping to pinpoint potential confounders that might influence the observed outcomes.

2.4 Preprocessing and Handling of Data

Categorical variables were encoded as binary indicators to facilitate their inclusion in regression analyses, while continuous variables were standardized to have a mean of zero and a standard deviation of one. These steps ensured comparability across variables measured on different scales.

2.5 Model Selection and Causal Inference

The causal effect of influenza vaccination on survival was estimated through a sequential approach involving multiple techniques designed to progressively address potential biases and confounding factors. The analysis began with the calculation of the Average Treatment Effect (ATE) using a naive approach, which simply subtracted the mean survival rate of vaccinated patients from that of unvaccinated patients without accounting for confounders. While this initial estimate provided a baseline measure of the effect of vaccination on survival, it did not account for the influence of other variables, limiting its reliability.

To refine the analysis, logistic regression model was employed to evaluate the effect of vaccination on survival. This model includes vaccination status as the primary factor, and incorporates a range of clinical and demographic characteristics, such as patient age, chronic conditions (e.g., cardiac, respiratory, neurological, renal, or liver diseases), a history of cancer or immunosuppression, sex, Indigenous Australian ethnicity, smoking habits, obesity, diabetes, functional status, the number of comorbidities, the hospital of admission, and the dominant influenza subtype (H3N2 or H1N1). This allowed for a more comprehensive assessment of the relationship between vaccination and survival while accounting for key confounders.

To address confounding more robustly, matching methods were applied. Both greedy matching (with and without replacement) and optimal matching were performed using the Mahalanobis distance metric, aiming to create balanced groups of vaccinated and unvaccinated patients based on their covariates. These covariates mirrored those used in the logistic regression models. The quality of the matches was evaluated using Standardized Mean Differences (SMD), with an SMD threshold of 0.1 considered indicative of adequate balance between groups. After matching, a randomization test was performed to assess the robustness of the matches, and logistic regression was re-applied to the matched dataset to re-evaluate the effect of vaccination on survival. Statistical significance was defined as a p-value < 0.05 .

As an alternative causal inference method, propensity score-based matching was also employed. Propensity scores, representing the probability of being vaccinated given the covariates, were calculated using a logistic regression model that included all potential confounders. Matching was then performed based on these scores to further improve balance between vaccinated and unvaccinated groups. Logistic regression models were again fitted to the propensity score-matched dataset to estimate the effect of vaccination on survival, and ATE values were calculated for both Mahalanobis-matched and propensity score-matched datasets, with statistical significance assessed at the 5% level.

Additionally, inverse probability of treatment weighting (IPTW) was applied as another robust causal inference method. IPTW assigns weights to individuals based on the inverse probability of their treatment assignment, as predicted by the propensity scores. Logistic regression models were then employed on the IPTW-adjusted dataset to assess the effect of vaccination on survival.

Beyond logistic regression, survival analysis was conducted to estimate the impact of vaccination on survival. Calendar years (2010–2017) were used as the time variable, with survival probability estimated across these years. Each calendar year was treated as a distinct survival interval, where patients who contracted influenza within that year formed a unique cohort. Survival time was thus measured as the proportion of patients within each year who survived versus died, providing year-by-year survival probabilities for vaccinated and non-vaccinated groups. Kaplan-Meier survival curves were constructed to estimate and compare survival probabilities between the two groups. The log-rank test was used to evaluate the statistical significance of differences between the curves. This analysis was performed both before adjustment, using the original dataset to explore crude survival differences, and after adjustment, using the matched dataset to account for confounders and derive adjusted survival probabilities.

To further investigate the association between vaccination and survival, Cox proportional hazards regression models were fitted. These models provided a more comprehensive assessment by estimating hazard ratios (HRs) and their 95% confidence intervals (CIs), quantifying the effect of vaccination on survival outcomes. Separate Cox models were

constructed for the original and matched datasets to assess the robustness of the findings under varying analytical frameworks.

Finally, the metrics obtained from these various approaches were compared to draw meaningful conclusions regarding the effectiveness of influenza vaccination in improving survival outcomes.

2.6 Subtype-Specific Analysis of Influenza Vaccination Effectiveness

To assess differences between the group affected by influenza during years when the dominant A subtype was H1N1 (2010, 2011, 2013, 2014, 2016) and years when the dominant A subtype was H3N2 (2012, 2015, 2017), the dataset was divided into two groups based on the dominant influenza subtype. For each group, propensity score matching was performed using the same methodologies outlined previously. Logistic regression analyses were then conducted for each matched dataset, and the Average Treatment Effect (ATE) was calculated. The results were compared to evaluate the effectiveness of vaccination in both scenarios, providing insights into the differential impact of vaccination across influenza subtypes.

3. Results

3.1 Descriptive Statistics

Before performing the matching, the analysis of the complete dataset revealed notable differences in survival rates between vaccinated and non-vaccinated individuals (Figure 1). Among the non-vaccinated group, 73.28% survived (628 individuals), while 26.72% (229 individuals) did not. Conversely, in the vaccinated group, the survival rate was lower, with 59.50% surviving (332 individuals) and 40.50% not surviving (226 individuals). When stratifying the data by influenza subtypes, these trends persisted. During H1N1-dominated seasons, the survival rate was 70.60% in non-vaccinated individuals and 55.51% in vaccinated individuals. Similarly, for H3N2-dominated seasons, the survival rate was 75.25% in non-vaccinated individuals compared to 62.42% in vaccinated individuals. These findings consistently indicated higher survival rates in the non-vaccinated group across the dataset and within influenza subtypes.

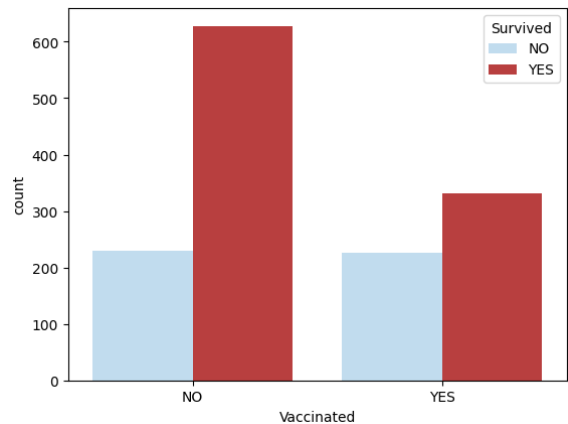


Figure 1. Distribution of Survival Outcomes Based on Vaccination Status.

A deeper examination of demographic and clinical characteristics highlighted important differences between the vaccinated and non-vaccinated groups. As shown in Table 1, vaccinated individuals were notably older, with a mean age of 51.1 years compared to 35.1 years in the non-vaccinated group. Certain clinical conditions were more prevalent in vaccinated individuals, such as chronic respiratory disease and chronic renal disease, suggesting targeted vaccination efforts for individuals with these conditions. However, chronic cardiac disease was slightly more common in non-vaccinated individuals. Other conditions, such as chronic neurological disease and malignancy, showed no significant differences in prevalence between groups. Vaccinated individuals were also more likely to have two or more comorbidities and less likely to have no comorbidities, further reflecting the focus on high-risk populations. Functional status also varied between groups, with vaccinated individuals more likely to be bedridden or significantly restricted in daily activities, while fewer vaccinated individuals had only mild restrictions. Vaccinated individuals were slightly less likely to be male and were underrepresented among Indigenous Australians. Smoking was less prevalent in the vaccinated group, while obesity rates were similar. Diabetes prevalence was marginally higher in vaccinated individuals.

Table 1. Baseline Characteristics of Vaccinated and Non-Vaccinated Individuals Across the Study Cohort.

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Variable	Overall (n=1415)	No Vaccinated (n=857)	Vaccinated (n=558)
Age, mean (SD)	41.4 (27.2)	35.1 (23.6)	51.1 (29.5)
Sex, n (%)	Female: 669 (47.3), Male: 746 (52.7)	Female: 395 (46.1), Male: 462 (53.9)	Female: 274 (49.1), Male: 284 (50.9)
Indigenous Australian, n (%)	No: 1321 (93.4) Yes: 94 (6.6)	No: 794 (92.6), Yes: 63 (7.4)	No: 527 (94.4), Yes: 31 (5.6)
Smoking, n (%)	No: 1224 (86.5), Yes: 191 (13.5)	No: 729 (85.1), Yes: 128 (14.9)	No: 495 (88.7), Yes: 63 (11.3)
Obesity, n (%)	No: 1030 (72.8), Yes: 385 (27.2)	No: 622 (72.6), Yes: 235 (27.4)	No: 408 (73.1), Yes: 150 (26.9)
Diabetes, n (%)	No: 1106 (78.2), Yes: 309 (21.8)	No: 673 (78.5), Yes: 184 (21.5)	No: 433 (77.6), Yes: 125 (22.4)
Functional Status, n (%)	Bed Bound: 26 (1.8) Limited: 145 (10.2) Mild Restriction: 247 (17.5) Not Restricted: 807 (57.0) Self-Caring: 190 (13.4)	Bed Bound: 12 (1.4) Limited: 83 (9.7) Mild Restriction: 159 (18.6) Not Restricted: 484 (56.5) Self-Caring: 119 (13.9)	Bed Bound: 14 (2.5) Limited: 62 (11.1) Mild Restriction: 88 (15.8) Not Restricted: 323 (57.9) Self-Caring: 71 (12.7)
Chronic Cardiac Disease, n (%)	No: 993 (70.2), Yes: 422 (29.8)	No: 589 (68.7), Yes: 268 (31.3)	No: 404 (72.4), Yes: 154 (27.6)
Chronic Respiratory Disease, n (%)	No: 1131 (79.9), Yes: 284 (20.1)	No: 716 (83.5), Yes: 141 (16.5)	No: 415 (74.4), Yes: 143 (25.6)
Chronic Neurological Disease, n (%)	No: 1161 (82.0), Yes: 254 (18.0)	No: 699 (81.6), Yes: 158 (18.4)	No: 462 (82.8), Yes: 96 (17.2)
Chronic Renal Disease, n (%)	No: 1371 (96.9), Yes: 44 (3.1)	No: 838 (97.8), Yes: 19 (2.2)	No: 533 (95.5), Yes: 25 (4.5)
Liver Disease, n (%)	No: 1356 (95.8), Yes: 59 (4.2)	No: 1356 (95.8), Yes: 59 (4.2)	No: 538 (96.4), Yes: 20 (3.6)
Malignancy, n (%)	No: 1303 (92.1), Yes: 112 (7.9)	No: 789 (92.1), Yes: 68 (7.9)	No: 514 (92.1), Yes: 44 (7.9)
Immunosuppression, n (%)	No: 1160 (82.0), Yes: 255 (18.0)	No: 705 (82.3), Yes: 152 (17.7)	No: 455 (81.5), Yes: 103 (18.5)
Number of Comorbidities, n (%)	0: 475 (33.6), 1: 437 (30.9), ≥2: 503 (35.5)	0: 326 (38.0), 1: 253 (29.5), ≥2: 278 (32.4)	0: 149 (26.7), 1: 184 (33.0), ≥2: 225 (40.3)

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The multivariate analysis further underscored the importance of age and other key variables in vaccination strategies and survival outcomes. Vaccinated individuals were significantly older than non-vaccinated ones ($p < 1.01 \times 10^{-25}$, 95% CI: -18.91 to -13.09), and survivors were significantly younger than non-survivors ($p < 1.40 \times 10^{-16}$, 95% CI: 10.38 to 16.66). Chronic respiratory disease ($p = 3.43 \times 10^{-5}$) and chronic renal disease ($p = 2.5 \times 10^{-2}$) were more prevalent among vaccinated individuals, reinforcing the targeted nature of vaccination programs. Significant associations were also observed for hospital ($p = 2.00 \times 10^{-2}$) and year of admission ($p = 1.00 \times 10^{-2}$), reflecting temporal and institutional differences in vaccination access and healthcare practices. The number of comorbidities was significantly higher in vaccinated individuals ($p = 4.20 \times 10^{-5}$), with more vaccinated individuals having multiple health conditions. However, no significant differences were observed for other clinical variables, such as chronic cardiac disease, liver disease, malignancy, immunosuppression, or functional status, suggesting either adequate balance between groups or limited influence of these factors on vaccination status. When examining the relationship between categorical variables and survival, chronic respiratory disease ($p = 1.65 \times 10^{-31}$) and chronic neurological disease ($p = 3.6 \times 10^{-2}$) were significantly associated with

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survival, aligning with their established roles as risk factors for severe influenza outcomes. The number of comorbidities ($p = 7.54 \times 10^{-5}$) and year of admission ($p = 1.6 \times 10^{-2}$) were also significant predictors, reflecting the cumulative risks posed by multiple health conditions and temporal variations in influenza severity and treatment practices.

Lastly, Table 2 highlights the differences in demographic and clinical characteristics during years dominated by H1N1 and H3N2 influenza subtypes, further contextualizing the observed survival trends.

Table 2. Baseline Characteristics Stratified by Dominant Influenza Subtypes (H1N1 vs. H3N2) and Vaccination Status.

Variable	H1N1 Overall (n=600)	H1N1 No Vaccinated (n=364)	H1N1 Vaccinated (n=236)	H3N2 Overall (n=815)	H3N2 No Vaccinated (n=493)	H3N2 Vaccinated (n=322)
Age, mean (SD)	42.3 (26.9)	35.8 (23.7)	52.3 (28.4)	40.7 (27.4)	34.5 (23.4)	50.2 (30.2)
Sex, n (%)	Female: 292 (48.7), Male: 308 (51.3)					
	Female: 173 (47.5), Male: 191 (52.5)					
	Female: 119 (50.4), Male: 117 (49.6)					
	Female: 377 (46.3), Male: 438 (53.7)					
	Female: 222 (45.0), Male: 271 (55.0)					
	Female: 155 (48.1), Male: 167 (51.9)					
Indigenous Australian, n (%)	No: 559 (93.2), Yes: 41 (6.8)	No: 336 (92.3), Yes: 28 (7.7)	No: 223 (94.5), Yes: 13 (5.5)	No: 762 (93.5), Yes: 53 (6.5)	No: 458 (92.9), Yes: 35 (7.1)	No: 304 (94.4), Yes: 18 (5.6)
Smoking, n (%)	No: 515 (85.8), Yes: 85 (14.2)	No: 302 (83.0), Yes: 62 (17.0)	No: 213 (90.3), Yes: 23 (9.7)	No: 709 (87.0), Yes: 106 (13.0)	No: 427 (86.6), Yes: 66 (13.4)	No: 282 (87.6), Yes: 40 (12.4)
Obesity, n (%)	No: 437 (72.8), Yes: 163 (27.2)	No: 270 (74.2), Yes: 94 (25.8)	No: 167 (70.8), Yes: 69 (29.2)	No: 593 (72.8), Yes: 222 (27.2)	No: 352 (71.4), Yes: 141 (28.6)	No: 241 (74.8), Yes: 81 (25.1)
Diabetes, n (%)	No: 473 (78.8), Yes: 127 (21.2)	No: 288 (79.1), Yes: 76 (20.9)	No: 185 (78.4), Yes: 51 (21.6)	No: 633 (77.7), Yes: 182 (22.3)	No: 385 (78.1), Yes: 108 (21.9)	No: 248 (77.0), Yes: 74 (23.0)
Functional Status, n (%)	Bed Bound: 11 (1.8), Limited: 58 (9.7), Mild Restriction: 104 (17.3), Not Restricted: 350 (58.3), Self-Caring: 77 (12.8)	Bed Bound: 7 (1.9), Limited: 30 (8.2), Mild Restriction: 69 (19.0), Not Restricted: 212 (58.2), Self-Caring: 46 (12.6)	Bed Bound: 4 (1.7), Limited: 28 (11.9), Mild Restriction: 35 (14.8), Not Restricted: 138 (58.5), Self-Caring: 31 (13.1)	Bed Bound: 15 (1.8), Limited: 87 (10.7), Mild Restriction: 143 (17.5), Not Restricted: 457 (56.1), Self-Caring: 113 (13.9)	Bed Bound: 5 (1.0), Limited: 53 (10.8), Mild Restriction: 90 (18.3), Not Restricted: 272 (55.2), Self-Caring: 73 (14.8)	Bed Bound: 10 (3.1), Limited: 34 (10.6), Mild Restriction: 53 (16.5), Not Restricted: 185 (57.5), Self-Caring: 40 (12.4)
Chronic Cardiac Disease, n (%)	No: 433 (72.2), Yes: 167 (27.8)	No: 258 (70.9), Yes: 106 (29.1)	No: 175 (74.2), Yes: 61 (25.8)	No: 560 (68.7), Yes: 255 (31.3)	No: 331 (67.1), Yes: 162 (32.9)	No: 229 (71.1), Yes: 93 (28.9)
Chronic Respiratory Disease, n (%)	No: 465 (77.5), Yes: 135 (22.5)	No: 293 (80.5), Yes: 71 (19.5)	No: 172 (72.9), Yes: 64 (27.1)	No: 666 (81.7), Yes: 149 (18.3)	No: 423 (85.8), Yes: 70 (14.2)	No: 243 (75.5), Yes: 79 (24.5)
Chronic Neurological Disease, n (%)	No: 496 (82.7), Yes: 104 (17.3)	No: 303 (83.2), Yes: 61 (16.8)	No: 193 (81.8), Yes: 43 (18.2)	No: 665 (81.6), Yes: 150 (18.4)	No: 396 (80.3), Yes: 97 (19.7)	No: 269 (83.5), Yes: 53 (16.5)
Chronic Renal Disease, n (%)	No: 583 (97.2), Yes: 17 (2.8)	No: 356 (97.8), Yes: 8 (2.2)	No: 227 (96.2), Yes: 9 (3.8)	No: 788 (96.7), Yes: 27 (3.3)	No: 482 (97.8), Yes: 11 (2.2)	No: 306 (95.0), Yes: 16 (5.0)
Liver Disease, n (%)	No: 575 (95.8), Yes: 25 (4.2)	No: 347 (95.3), Yes: 17 (4.7)	No: 228 (96.6), Yes: 8 (3.4)	No: 781 (95.8), Yes: 34 (4.2)	No: 471 (95.5), Yes: 22 (4.5)	No: 310 (96.3), Yes: 12 (3.7)
Malignancy, n (%)	No: 548 (91.3), Yes: 52 (8.7)	No: 335 (92.0), Yes: 29 (8.0)	No: 213 (90.3), Yes: 23 (9.7)	No: 755 (92.6), Yes: 60 (7.4)	No: 454 (92.1), Yes: 39 (7.9)	No: 301 (93.5), Yes: 21 (6.5)
Immunosuppression, n (%)	No: 493 (82.2), Yes: 107 (17.8)	No: 299 (82.1), Yes: 65 (17.9)	No: 194 (82.2), Yes: 42 (17.8)	No: 667 (81.8), Yes: 148 (18.2)	No: 406 (82.4), Yes: 87 (17.6)	No: 261 (81.1), Yes: 61 (18.9)
Number of Comorbidities, n (%)	0: 211 (35.2), 1: 171 (28.5), ≥2: 218 (36.3)	0: 146 (40.1), 1: 104 (28.6), ≥2: 114 (31.3)	0: 65 (27.5), 1: 67 (28.4), ≥2: 104 (44.1)	0: 264 (32.4), 1: 266 (32.6), ≥2: 285 (35.0)	0: 180 (36.5), 1: 149 (30.2), ≥2: 164 (33.3)	0: 84 (26.1), 1: 117 (36.3), ≥2: 121 (37.6)

3.2 Causal Inference and Logistic regression models

The calculation of the naive Average Treatment Effect (ATE) yielded a value of -0.1378, suggesting a 13.8% lower survival rate among vaccinated individuals compared to unvaccinated ones. However, this result is influenced by bias due to unbalanced confounders between the two groups, rendering the estimate unreliable as it does not account for these discrepancies.

The ATE calculated using logistic regression, which included vaccination status as well as all potential confounders as covariates but without balancing these confounders between groups, resulted in a value of 0.0381. This suggests a modest 3.8% increase in survival among vaccinated individuals compared to unvaccinated ones; however, the lack of balanced groups limits the reliability of this estimate. Additionally, the coefficient for the covariate vaccinated in the model was -0.2107, with a p-value of 0.122, indicating that the effect of vaccination on survival was not statistically significant in this model.

The Greedy matching with Mahalanobis distance, allowing replacement, improved the balance between vaccinated and unvaccinated groups for many variables. However, some residual imbalances persisted. Age remained notably imbalanced (SMD: 0.6), with vaccinated individuals still older on average. Similarly, chronic respiratory disease exhibited moderate imbalance (SMD: 0.227), reflecting a higher prevalence in vaccinated individuals, and chronic renal disease showed a mild imbalance (SMD: 0.126). Among comorbidity counts, individuals with two or more comorbidities also retained a significant imbalance (SMD: 0.164). Most other clinical and demographic variables achieved adequate balance, including chronic cardiac disease (SMD: 0.081), liver disease (SMD: 0.049), and smoking status (SMD: 0.108). Functional status variables were generally well matched, though mild residual imbalances were observed for mild restriction (SMD: 0.074) and limited functional status (SMD: 0.047). Hospital-related variables varied in balance, with Hospital 12 showing a significant imbalance (SMD: 0.223), while others such as Hospital 11 (SMD: 0.008) achieved strong balance. The distribution of influenza subtypes (H3N2 vs. others) was well balanced, with an SMD of 0.004. Despite the overall improvement in balance, age, chronic respiratory disease, and individuals with higher comorbidity counts remained areas of concern, indicating potential bias that may affect subsequent analyses.

The Greedy matching without replacement retained only 52% of the dataset, leading to improved balance for several variables compared to the with-replacement match. Age achieved a lower SMD of 0.469 (vs. 0.6 with replacement), and imbalances in chronic respiratory disease (SMD: 0.161) and chronic neurological disease (SMD: 0.144) were reduced. Complete balance was achieved for chronic renal disease and specific hospital-related variables (e.g., Hospital 4 and Hospital 6, SMD: <0.001). Functional status categories also showed slight improvements in balance. However, some variables, such as indigenous status (SMD: 0.147) and diabetes (SMD: 0.152), remained moderately imbalanced. Residual discrepancies were noted for number of comorbidities with one condition (SMD: 0.134) and functional status: not restricted (SMD: 0.227). While the stricter matching criteria improved balance for critical variables like age, it resulted in greater data loss and maintained imbalances for some subgroups, highlighting the trade-offs inherent in the matching approach.

The optimal matching approach retained 79% of the original dataset, offering better data retention than the Greedy matching without replacement but slightly less than the with-replacement method. While balance improved for many variables, notable imbalances remained. Age showed a significant SMD of 0.556, indicating vaccinated individuals were still older on average, and chronic respiratory disease exhibited a moderate imbalance (SMD: 0.230), with a higher prevalence among vaccinated individuals. Other clinical variables, such as chronic cardiac disease (SMD: 0.02), liver disease (SMD: 0.03), and smoking status (SMD: 0.055), achieved good balance, as did most hospital-related variables, with SMDs generally below 0.05. Functional status categories showed mild residual imbalances, particularly for limited functional status (SMD: 0.072) and mild restriction (SMD: 0.035). The number of comorbidities retained moderate imbalances, with SMDs of 0.113 for one comorbidity and 0.115 for two or more. Despite these efforts, the calculated ATE after matching was -0.127, indicating a 12.7% lower survival rate among vaccinated individuals compared to unvaccinated ones. This suggests residual confounding, particularly from variables like age and chronic respiratory disease, which could still bias the results and warrant further adjustments or analyses to ensure robustness.

Performing the Randomization test with 1,000 permutations yielded a high p-value, indicating no significant evidence of differences between the vaccinated and unvaccinated groups, even after the matching process. This result suggests that any observed differences in outcomes between the groups are likely due to residual confounding or random variation rather than systematic differences attributable to vaccination status.

After optimal matching, a logistic regression analysis revealed a positive association between influenza vaccination and survival, with an estimated effect size of 0.3846 ($p < 0.001$). The estimated ATE from this analysis was 0.09498, indicating a 9.5% increase in survival among vaccinated individuals. However, given the inconclusive result of the Randomization test, which showed no evidence of systematic differences between vaccinated and unvaccinated groups, this estimate cannot be considered reliable. The observed association is likely influenced by residual confounding or random variation, and caution is needed when interpreting these results.

Given the limited efficacy of previously employed matching methods in achieving balance between vaccinated and non-vaccinated groups, an alternative approach was undertaken using propensity score matching. Following this matching process, a logistic regression model was applied to predict survival as a function of vaccination status.

The results of the analysis revealed an Average Treatment Effect (ATE) of -0.011 (SE = 0.044, $p = 0.805$, 95% CI: -0.098 to 0.076). This estimate indicates no statistically significant difference in survival outcomes between vaccinated and non-vaccinated individuals. The near-zero ATE, along with the wide confidence interval and high p -value, suggests that vaccination did not have a discernible impact on survival in the matched dataset.

Applying IPTW resulted in a significant association between vaccination status and survival ($p < 0.001$). The odds ratio for survival among vaccinated individuals was 0.536 (95% CI: 0.397–0.851), indicating 46% lower odds of survival compared to non-vaccinated individuals. These results contrast with the findings from matching methods, highlighting the influence of analytical approaches on observed effects.

3.3 Causal Inference and Survival regression models

The Kaplan-Meier survival analysis revealed significant differences in survival probabilities between vaccinated and non-vaccinated groups, both before and after adjusting for confounders (Figure 2). The log-rank test on the original dataset identified a highly significant disparity between the groups ($p < 2.31 \times 10^{-8}$), which persisted after adjustment using the matched dataset ($p < 3.12 \times 10^{-6}$).

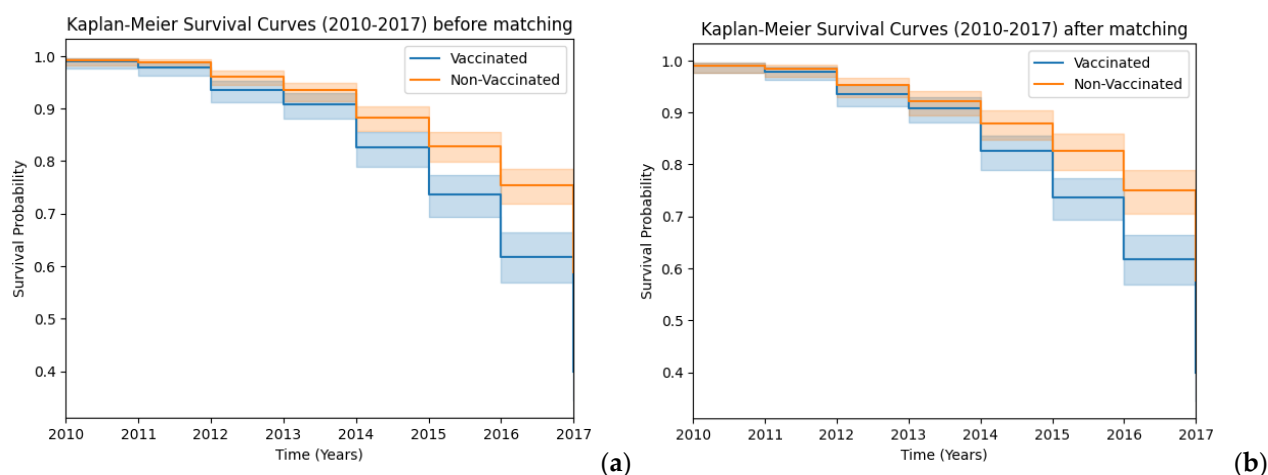


Figure 2. (a) Kaplan-Meier survival curves before matching for vaccinated and unvaccinated groups, showing survival probabilities over time. (b) Kaplan-Meier survival curves after matching, illustrating survival probabilities after adjusting for confounding factors.

In the unadjusted analysis, the Cox proportional hazards model showed that vaccination was associated with an increased hazard of death (HR = 1.69, 95% CI: 1.41–2.03, $p < 0.005$), indicating a higher likelihood of death among vaccinated individuals compared to non-vaccinated ones. After the Optimal matching, the adjusted Cox model produced a slightly lower hazard ratio of 1.63 (95% CI: 1.32–2.00, $p < 0.005$), suggesting that vaccination remained associated with an increased risk of death, although the effect size was reduced.

3.4 Subtype-Specific Analysis of Influenza Vaccination Effectiveness

For seasons dominated by H1N1, the estimated ATE was -0.135 (95% CI: -0.404 to 0.134, $p = 0.326$), indicating no statistically significant effect of vaccination on survival. While the negative point estimate suggests a possible trend toward slightly worse survival outcomes for vaccinated individuals, the wide confidence interval and lack of statistical significance underscore the uncertainty of this result. During H3N2-dominated years, the estimated ATE was 0.009

(95% CI: -0.102 to 0.120, $p = 0.871$), similarly demonstrating no significant effect of vaccination on survival. The near-zero-point estimate and the confidence interval both suggest no meaningful difference in survival outcomes between vaccinated and unvaccinated individuals in these seasons, reinforcing the lack of evidence for a vaccine effect in these contexts.

4. Discussion and Conclusions

This study aimed to evaluate the impact of influenza vaccination on reducing mortality among hospitalized patients in Australia between 2010 and 2017, leveraging a causal inference framework to address confounding and assess vaccine effectiveness (VE) under varying conditions. Our findings reveal a complex and nuanced relationship between vaccination and survival, underscoring the challenges of disentangling causal effects from confounding factors.

Initially, crude analyses suggested a counterintuitive association between vaccination and increased mortality, with vaccinated individuals exhibiting significantly lower survival rates compared to their non-vaccinated counterparts. This apparent paradox reflects the influence of confounding by indication, where vaccination is disproportionately administered to individuals with higher baseline risks, such as older adults and those with multiple comorbidities. These characteristics, intrinsic to high-risk groups, likely masked the protective effects of vaccination in the unadjusted analysis.

After accounting for confounders through propensity score matching and other statistical adjustments, the observed association between vaccination and mortality became neutral, with no statistically significant difference in survival between vaccinated and non-vaccinated individuals. This shift highlights the critical role of confounding in driving the initial negative association and demonstrates the importance of employing robust causal inference methods to avoid biased estimates. However, even after these adjustments, the vaccination effect did not turn positive, failing to show a protective impact on survival in the adjusted models.

Subtype-specific analyses further revealed no significant differences in survival outcomes during H1N1- or H3N2-dominated seasons. While the ATE estimates suggested slight trends, such as a potential negative effect in H1N1 years and a near-neutral effect in H3N2 years, none were statistically significant. This lack of effect may be influenced by residual confounding, unmeasured variables, or vaccine efficacy limitations specific to certain subtypes and patient populations.

Several limitations must be considered when interpreting these findings. Although efforts were made to adjust for confounding, residual confounding from unmeasured factors, such as frailty, prior vaccination history, or timing of vaccination, may have influenced the results. Additionally, the dataset included hospitalized patients, a population already at high risk for severe outcomes, which may limit the generalizability of these findings to broader community settings. Methodological challenges, such as the potential for selection bias and the reliance on observational data, further underscore the need for cautious interpretation of the results. Subtype-specific analyses, while valuable for understanding the heterogeneity of vaccine impact, were limited by sample size constraints, particularly in stratified analyses.

Despite these limitations, the findings provide critical insights for influenza vaccination strategies. The observed neutral effect after adjustment emphasizes the need to focus on improving vaccine effectiveness, particularly among high-risk populations. Strategies such as enhancing vaccine-virus antigenic match, developing more immunogenic vaccines for older adults, and addressing barriers to timely vaccination could help mitigate the observed disparities in survival outcomes. Furthermore, efforts to expand vaccination coverage to underrepresented populations, such as Indigenous Australians and individuals with limited healthcare access, remain a public health priority.

In conclusion, while this study underscores the significant impact of confounding on the apparent relationship between influenza vaccination and mortality, it also reveals a neutral effect of vaccination after adjustment, suggesting that the potential benefits of vaccination on survival remain unproven in this context. Future research should prioritize randomized controlled trials or alternative designs that minimize confounding, alongside continued efforts to develop more effective vaccines. By addressing these challenges, vaccination programs can better achieve their goal of reducing the global burden of seasonal influenza and its most severe outcomes.

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